

Qatar-1b

R-0365

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_180602 · created 2026-07-18T14:27:23+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458271.8368 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.016
Transit depth (Rp/Rs)^2	3.43 +/- 0.61 [%]
Orbital Inclination (inc)	81.76 +/- 0.46
Airmass coefficient 1 (a1)	1.008 +/- 0.011
Airmass coefficient 2 (a2)	-0.0051 +/- 0.0095
Scatter in the residuals of the lightcurve fit is	1.09 %
Best Comparison Star	#3 - [516, 193]
Optimal Aperture	2.95
Optimal Annulus	9.68
Transit Duration (day)	0.0569 +/- 0.0048

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.185 $\pm$ 0.016 vs 0.1463 (deviation 2.4 σ)
Duration fit vs published	0.0569 d vs 0.0692 d (deviation 2.54 σ)
Mid-time O-C	-2.0 min
Depth SNR	11.5
Residual scatter	1.09 %

Light curve

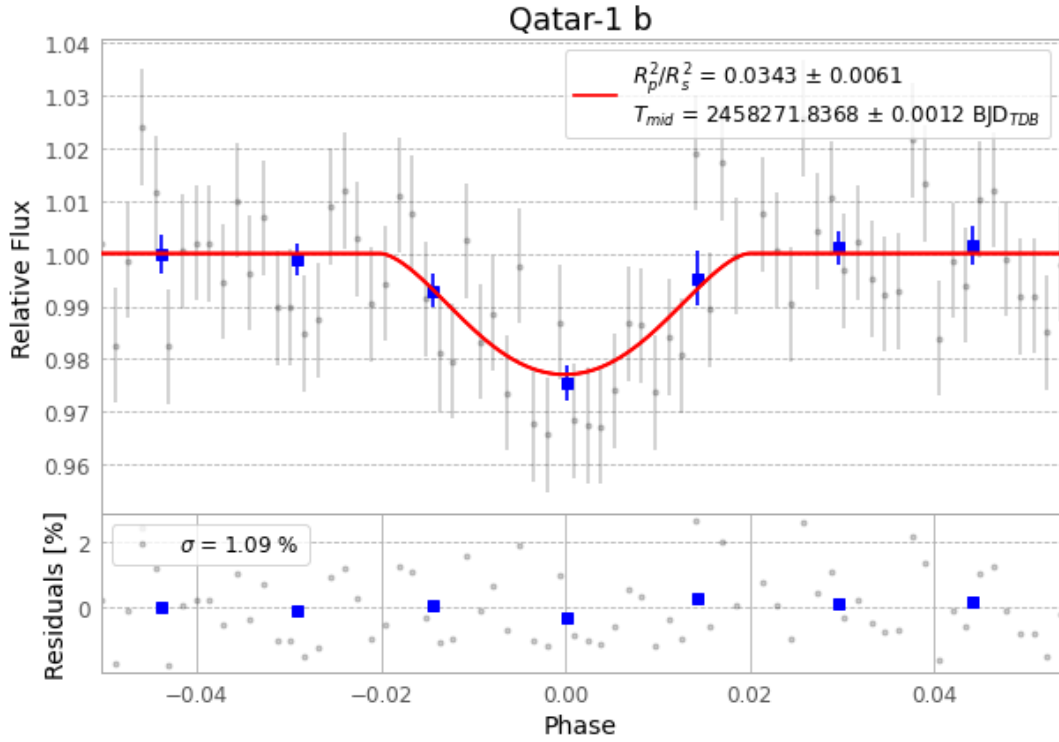


Figure 1 — FinalLightCurve_Qatar-1 b_2018-06-01.png (SHA-256 2f0c04d098fd654a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [468, 250], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 85f9658b3132f224...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 9e597cf

Data provenance

72 FITS frames (47 MB), Qatar-1180602062112.FITS → Qatar-1180602095412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-180602.log (425d4eea83c4...), FinalLightCurve_Qatar-1 b_2018-06-01.png (2f0c04d098fd...), FinalLightCurve_Qatar-1 b_2018-06-01.csv (ccc065f4e725...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 14:27Z.